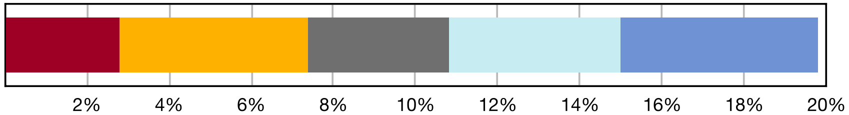
ADVANCED
Stage 6

Calculus (Adv), C3 Applications of Calculus (Adv)
Maxima and Minima (Y12)

Teacher: SHS
Exam Equivalent Time: 79.5 minutes *(based on allocation of 1.5 minutes per mark)*

C3 Applications of Calculus



- The Derivative Function and its Graph
- Curve Sketching
- Tangents
- Maxima and Minima
- Rates of Change

***SmarterMaths analytics based on the average contribution to new syllabus Advanced Maths exams since 2020.**

HISTORICAL CONTRIBUTION

- *C3 Applications of Calculus* is the 800 pound gorilla in the Advanced rainforest, contributing a massive 19.8% to new syllabus exams since introduced in 2020.
- This topic has been split into five sub-topics for analysis purposes: *1-The Derivative Function and its Graph (2.8%), 2-Curve Sketching (4.6%), 3-Tangents (3.4%), 4-Maxima and Minima (4.2%) and 5-Rates of Change (4.8%).*
- This analysis looks at *Maxima and Minima.*

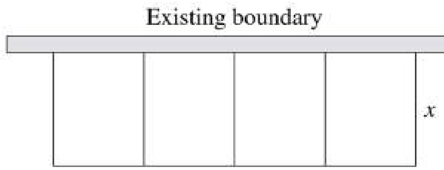
HSC ANALYSIS - What to expect and common pitfalls

- *Maxima and Minima (4.2%)* appears in every exam with significant mark allocations almost guaranteed. Its importance in achieving a band 5-6 result cannot be overstated.
- *Max/Min* problems will typically ask students to prove an equation in the earlier parts. A band 4 level task that presents a good opportunity for all students to score well.
- HSC Markers remind students to work with the given equation of the question, *even if they couldn't successfully complete the proof in the earlier part.*
- *Max/Min* problems have had a number of specific themes. *Area/Volume equations* have dominated, appearing in 4 out of 5 new syllabus exams. Other themes include *distance* (2017 and 2013) and a *cost equation* (2009).
- *Max/Min* was novelly examined within the *Probability Density Function* topic in 2021 and is included in the database under that topic.

Questions

1. Calculus, 2ADV C3 2016 HSC 14c

A farmer wishes to make a rectangular enclosure of area 720 m². She uses an existing straight boundary as one side of the enclosure. She uses wire fencing for the remaining three sides and also to divide the enclosure into four equal rectangular areas of width x m as shown.



i. Show that the total length, l m, of the wire fencing is given by

$$l = 5x + \frac{720}{x}. \quad \textit{(1 mark)}$$

ii. Find the minimum length of wire fencing required, showing why this is the minimum length. ***(3 marks)***

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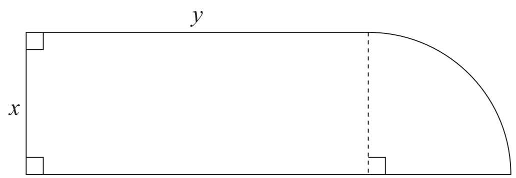
2. Calculus, 2ADV C3 2020 HSC 25

A landscape gardener wants to build a garden in the shape of a rectangle attached to a quarter-circle. Let x and y be the dimensions of the rectangle in metres, as shown in the diagram.



The garden bed is required to have an area of 36 m² and to have a perimeter which is as small as possible. Let P metres be the perimeter of the garden bed.

a. Show that $P = 2x + \dfrac{72}{x}$. ***(3 marks)***

b. Find the smallest possible perimeter of the garden bed, showing why this is the minimum perimeter. ***(4 marks)***

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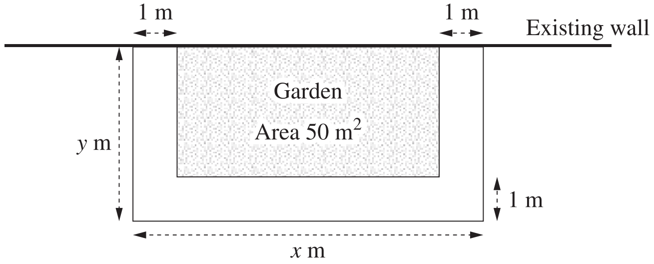
3. Calculus, 2ADV C3 2023 HSC 24

A gardener wants to build a rectangular garden of area 50 m² against an existing wall as shown in the diagram. A concrete path of width 1 metre is to be built around the other three sides of the garden.



Let x and y be the dimensions, in metres, of the outer rectangle as shown.

a. Show that $y = \dfrac{50}{x-2} + 1$. ***(1 mark)***

b. Find the value of x such that the area of the concrete path is a minimum. Show that your answer gives a minimum area. ***(4 marks)***

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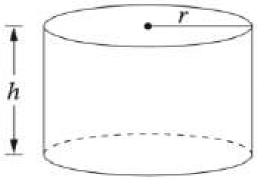
4. Calculus, 2ADV C3 2010 HSC 5a

A rainwater tank is to be designed in the shape of a cylinder with radius r metres and height h metres.



The volume of the tank is to be 10 cubic metres. Let A be the surface area of the tank, including its top and base, in square metres.

i. Given that $A = 2\pi r^2 + 2\pi r h$, show that $A = 2\pi r^2 + \dfrac{20}{r}$. ***(2 marks)***

ii. Show that A has a minimum value and find the value of r for which the minimum occurs. ***(3 marks)***

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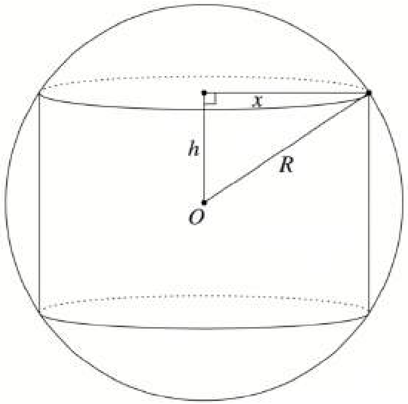
5. Calculus, 2ADV C3 2005 HSC 8a



A cylinder of radius x and height $2h$ is to be inscribed in a sphere of radius R centred at O as shown.

i. Show that the volume of the cylinder is given by

$$V = 2\pi h\left(R^2 - h^2\right).$$ ***(1 mark)***

ii. Hence, or otherwise, show that the cylinder has a maximum volume when $h = \dfrac{R}{\sqrt{3}}$. ***(3 marks)***

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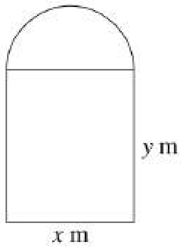
6. Calculus, 2ADV C3 2014 HSC 16c

The diagram shows a window consisting of two sections. The top section is a semicircle of diameter x m. The bottom section is a rectangle of width x m and height y m.

The entire frame of the window, including the piece that separates the two sections, is made using 10 m of thin metal.



The semicircular section is made of coloured glass and the rectangular section is made of clear glass.

Under test conditions the amount of light coming through one square metre of the coloured glass is 1 unit and the amount of light coming through one square metre of the clear glass is 3 units.

The total amount of light coming through the window under test conditions is L units.

i. Show that $y = 5 - x\left(1 + \frac{\pi}{4}\right)$. *(2 marks)*

ii. Show that $L = 15x - x^2\left(3 + \frac{5\pi}{8}\right)$. *(2 marks)*

iii. Find the values of x and y that maximise the amount of light coming through the window under test conditions. *(3 marks)*

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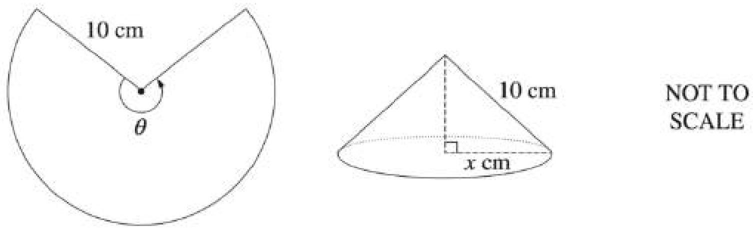
7. Calculus, 2ADV C3 2018 HSC 16a

A sector with radius 10 cm and angle θ is used to form the curved surface of a cone with base radius x cm, as shown in the diagram.



The volume of a cone of radius r and height h is given by $V = \frac{1}{3}\pi r^2 h$.

i. Show that the volume, V cm³, of the cone described above is given by

$$V = \frac{1}{3}\pi x^2\sqrt{100 - x^2}.$$ *(1 mark)*

ii. Show that $\frac{dV}{dx} = \frac{\pi x\left(200 - 3x^2\right)}{3\sqrt{100 - x^2}}$. *(2 marks)*

iii. Find the exact value of θ for which V is a maximum. *(3 marks)*

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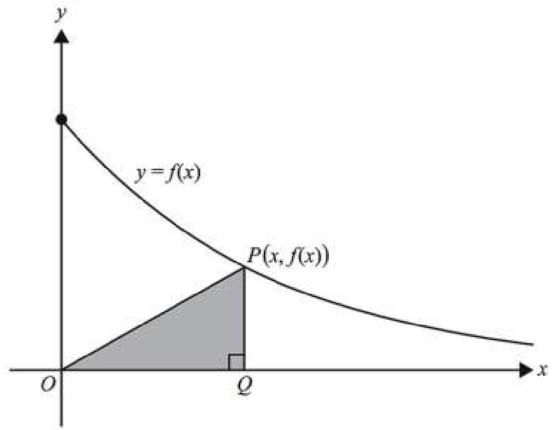
8. Calculus, 2ADV C4 SM-Bank 12

Let $f(x) = 2e^{-\frac{x}{5}}$ for $x \geq 0$

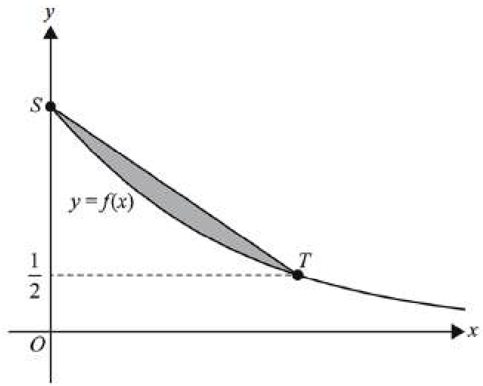
A right-angled triangle OQP has vertex O at the origin, vertex Q on the x-axis and vertex P on the graph of f, as shown. The coordinates of P are $(x, f(x))$.



i. Find the area, A, of the triangle OPQ in terms of x. ***(1 mark)***

ii. Find the maximum area of triangle OQP and the value of x for which the maximum occurs. ***(3 marks)***

iii. Let S be the point on the graph of f on the y-axis and let T be the point on the graph of f with the y-coordinate $\frac{1}{2}$. Find the area of the region bounded by the graph of f and the line segment ST. ***(2 marks)***



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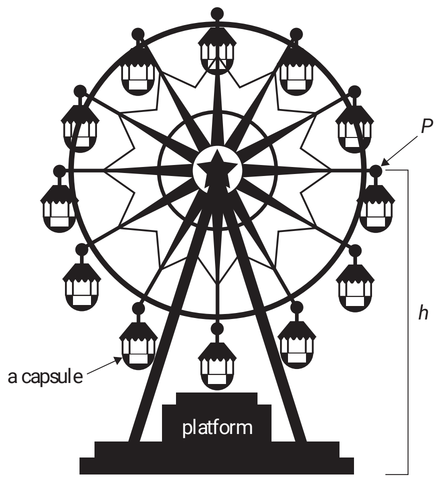
9. Trigonometry, 2ADV T3 SM-Bank 16

Sammy visits a giant Ferris wheel. Sammy enters a capsule on the Ferris wheel from a platform above the ground. The Ferris wheel is rotating anticlockwise. The capsule is attached to the Ferris wheel at point P.

The height of P above the ground, h, is modelled by $h(t) = 65 - 55\cos\left(\frac{\pi t}{15}\right)$, where t is the time in minutes after Sammy enters the capsule and h is measured in metres.

Sammy exits the capsule after one complete rotation of the Ferris wheel.



i. State the minimum and maximum heights of P above the ground. ***(1 mark)***

ii. For how much time is Sammy in the capsule? ***(1 mark)***

iii. Find the rate of change of h with respect to t and, hence, state the value of t at which the rate of change of h is at its maximum. ***(2 marks)***

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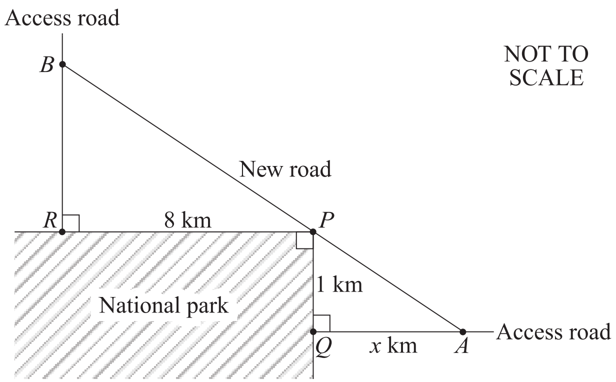
10. Calculus, 2ADV C3 2019 HSC 15c

The entry points, R and Q, to a national park can be reached via two straight access roads. The access roads meet the national park boundaries at right angles. The corner, P, of the national park is 8 km from R and 1 km from Q. The boundaries of the national park form a right angle at P.

A new straight road is to be built joining these roads and passing through P.

Points A and B on the access roads are to be chosen to minimise the distance, D km, from A to B along the new road.

Let the distance QA be x km.



i. Show that $D^2 = (x+8)^2 + \left(\frac{8}{x}+1\right)^2$. *(3 marks)*

ii. Show that $x = 2$ gives the minimum value of D^2. *(3 marks)*

Worked Solutions

1. Calculus, 2ADV C3 2016 HSC 14c

i. $\text{Area} = xy$

$$720 = xy$$

$$y = \frac{720}{x}$$

$$l = 5x + y$$

$$= 5x + \frac{720}{x} \quad \ldots \text{as required}$$

ii. $\dfrac{dl}{dx} = 5 - \dfrac{720}{x^2}$

$$\frac{d^2l}{dx^2} = \frac{1440}{x^3}$$

Max/Min when $\dfrac{dl}{dx} = 0$,

$$5 = \frac{720}{x^2}$$

$$x^2 = 144$$

$$x = 12, \quad x > 0$$

When $x = 12$, $\dfrac{d^2l}{dx^2} > 0$

$\therefore$ Minimum occurs when $x = 12$

$\therefore$ Minimum fencing

$$= 5 \times 12 + \frac{720}{12}$$

$$= 120 \text{ m}$$

2. Calculus, 2ADV C3 2020 HSC 25

a. $\text{Area} = xy + \dfrac{1}{4}\pi r^2$

$$36 = xy + \dfrac{1}{4}\pi x^2$$

$$xy = 36 - \dfrac{\pi x^2}{4}$$

$$y = \dfrac{36}{x} - \dfrac{\pi x}{4}$$

♦ *Mean mark part (a) 47%.*

$$\therefore P = 2x + 2y + \dfrac{1}{4}(2\pi x)$$

$$= 2x + 2\left(\dfrac{36}{x} - \dfrac{\pi x}{4}\right) + \dfrac{\pi x}{2}$$

$$= 2x + \dfrac{72}{x} - \dfrac{\pi x}{2} + \dfrac{\pi x}{2}$$

$$= 2x + \dfrac{72}{x}$$

b. $P = 2x + \dfrac{72}{x}$

$$\dfrac{dP}{dx} = 2 - 72x^{-2}$$

$$\dfrac{d^2P}{dx^2} = 144x^{-3}$$

Max or min when $\dfrac{dP}{dx} = 0$:

$$2 - \dfrac{72}{x^2} = 0$$

$$2x^2 = 72$$

$$x^2 = 36$$

$$x = 6, \quad (x > 0)$$

When $x = 6$,

$$\dfrac{d^2P}{dx^2} = \dfrac{144}{6^3} = \dfrac{2}{3} > 0$$

$\Rightarrow$ MIN at $x = 6$

$$\therefore P_{\text{min}} = 2 \times 6 + \dfrac{72}{6}$$

$$= 24 \text{ m}$$

3. Calculus, 2ADV C3 2023 HSC 24

a. $(y-1)(x-2) = 50$

$$yx - 2y - x + 2 = 50$$

$$y(x-2) = 48 + x$$

$$y = \dfrac{48 + x}{x - 2}$$

$$y = \dfrac{50 + (x-2)}{x - 2}$$

$$y = \dfrac{50}{x-2} + 1$$

♦♦ Mean mark (b) 33%.

b. Let $A =$ area of concrete path

$$A = xy - 50$$

$$= x\left(\dfrac{50}{x-2} + 1\right) - 50$$

$$= \dfrac{50x}{x-2} + x - 50$$

$$\dfrac{dA}{dx} = \dfrac{50(x-2) - 50x}{(x-2)^2} + 1$$

$$= \dfrac{-100}{(x-2)^2} + 1$$

Max/min when $\dfrac{dA}{dx} = 0$

$$\dfrac{-100}{(x-2)^2} + 1 = 0$$

$$(x-2)^2 = 100$$

$$x - 2 = \pm 10$$

$$x = 12 \ (x > 0)$$

Check gradients about $x = 12$:

x	10	12	14
$\dfrac{dA}{dx}$	$-\frac{9}{16}$	0	$\frac{11}{36}$
	\	_	/

$\therefore$ Minimum at $x = 12$.

4. Calculus, 2ADV C3 2010 HSC 5a

i. Show $A = 2\pi r^2 + \dfrac{20}{r}$

Since $V = \pi r^2 h = 10 \quad \Rightarrow \; h = \dfrac{10}{\pi r^2}$

Substituting into A

$$A = \pi r^2 + 2\pi r\left(\frac{10}{\pi r^2}\right)$$

$$= 2\pi r^2 + \frac{20}{r} \quad \ldots \text{ as required}$$

MARKER'S COMMENT: Students MUST know the volume formula for a cylinder. Those that did and stated $10 = \pi r^2 h$ most often completed this question efficiently.

ii.

$$A = 2\pi r^2 + \frac{20}{r}$$

$$\frac{dA}{dr} = 4\pi r - \frac{20}{r^2}$$

$$\frac{d^2A}{dr^2} = 4\pi + \frac{40}{r^3} > 0 \qquad (r > 0)$$

$\Rightarrow$ MIN occurs when $\dfrac{dA}{dr} = 0$

$$4\pi r - \frac{20}{r^2} = 0$$

$$4\pi r^3 - 20 = 0$$

$$4\pi r^3 = 20$$

$$r = \sqrt[3]{\frac{5}{\pi}}$$

$$= 1.16754\ldots$$

$$= 1.17 \text{ metres } \; (2 \text{ d.p.})$$

♦ *Mean mark 44%*

MARKER'S COMMENT: The "table method" or 1st derivative test for proving a minimum (i.e. showing how $\frac{dA}{dr}$ changes sign) was also quite successful.

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5. Calculus, 2ADV C3 2005 HSC 8a

i. $V = \pi r^2 h \quad$ (general form)

$$= \pi x^2 \times 2h$$

Using Pythagoras

$$R^2 = h^2 + x^2$$

$$x^2 = R^2 - h^2$$

$$\therefore V = 2\pi h\left(R^2 - h^2\right) \quad \ldots \text{as required.}$$

ii.

$$V = 2\pi h\left(R^2 - h^2\right)$$

$$= 2\pi R^2 h - 2\pi h^3$$

$$\frac{dV}{dh} = 2\pi R^2 - 3 \times 2\pi h^2$$

$$= 2\pi R^2 - 6\pi h^2$$

$$\frac{d^2V}{dh^2} = -\,12\pi h$$

Max or min when $\dfrac{dV}{dh} = 0$

$$2\pi R^2 - 6\pi h^2 = 0$$

$$6\pi h^2 = 2\pi R^2$$

$$h^2 = \frac{R^2}{3}$$

$$h = \frac{R}{\sqrt{3}}, \quad h > 0$$

When $h = \dfrac{R}{\sqrt{3}}$

$$\frac{d^2V}{dh^2} = -\,12\pi \times \frac{R}{\sqrt{3}} < 0$$

$\therefore$ Volume is a maximum when $h = \dfrac{R}{\sqrt{3}}$ units.

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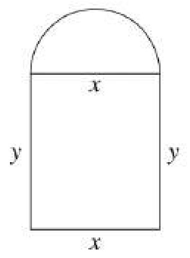
6. Calculus, 2ADV C3 2014 HSC 16c

i.



Frame is 10m

$$10 = 2x + 2y + \frac{1}{2}\pi x$$

$$2y = 10 - 2x - \frac{1}{2}\pi x$$

$$\therefore y = 5 - x - \frac{\pi}{4}x$$

$$= 5 - x\left(1 + \frac{\pi}{4}\right) \quad \text{. . . as required.}$$

ii. Area (clear) $= x \times y$

♦ *Mean mark 35%*

$$\text{Area (colour)} = \frac{1}{2} \times \pi r^2$$

$$= \frac{1}{2} \times \pi\left(\frac{x}{2}\right)^2$$

$$= \frac{\pi x^2}{8}$$

$$\therefore\ L = 3xy + \left(\frac{\pi x^2}{8} \times 1\right)$$

$$= 3x\left[5 - x\left(1 + \frac{\pi}{4}\right)\right] + \frac{\pi x^2}{8} \quad \text{(see part (i))}$$

$$= 15x - 3x^2 - \frac{3x^2\pi}{4} + \frac{x^2\pi}{8}$$

$$= 15x - 3x^2 - \frac{5x^2\pi}{8}$$

$$= 15x - x^2\left(3 + \frac{5\pi}{8}\right) \quad \text{. . . as required}$$

iii.

$$L = 15x - x^2\left(3 + \frac{5\pi}{8}\right)$$

$$\frac{dL}{dx} = 15 - 2x\left(3 + \frac{5\pi}{8}\right)$$

♦ *Mean mark 38%*
COMMENT: A sanity check for your answer could be to compare your answers to the perimeter restriction of 10m.

$$\frac{d^2L}{dx^2} = -2\left(3+\frac{5\pi}{8}\right)$$

Max or min when $\dfrac{dL}{dx} = 0$

$$15 - 2x\left(3x+\frac{5\pi}{8}\right) = 0$$

$$2x\left(3+\frac{5\pi}{8}\right) = 15$$

$$x = \frac{15}{2\left(3+\frac{5\pi}{8}\right)}$$

$$= 1.51103\ldots$$

$$= 1.511 \quad (3 \text{ d.p.})$$

Since $\dfrac{d^2L}{dx^2} < 0 \quad \Rightarrow \text{MAX}$

When $x = 1.511$

$$y = 5 - 1.511\left(1+\frac{\pi}{4}\right)$$

$$= 2.3022\ldots$$

$$= 2.302 \ (3 \text{ d.p.})$$

$\therefore$ MAX light when $x = 1.511$ m
and $y = 2.302$ m.

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7. Calculus, 2ADV C3 2018 HSC 16a

i. $V = \dfrac{1}{3}\pi r^2 h$

Using Pythagoras,

$h = \sqrt{100 - x^2}$

$r = x$

$\therefore$ Volume $= \dfrac{1}{3}\pi x^2\sqrt{100-x^2}$

ii.

♦ Mean mark 45%.

$$V = \frac{1}{3}\pi x^2\sqrt{100-x^2}$$

$$\frac{dV}{dh} = \frac{1}{3}\pi\left[2x \cdot \sqrt{100-x^2} - 2x \cdot \frac{1}{2}\left(100-x^2\right)^{-\frac{1}{2}} \cdot x^2\right]$$

$$= \frac{1}{3}\pi\left[\frac{2x\left(100-x^2\right)-x^3}{\sqrt{100-x^2}}\right]$$

$$= \frac{1}{3}\pi\left[\frac{200x-2x^3-x^3}{\sqrt{100-x^2}}\right]$$

$$= \frac{\pi x\left(200-3x^2\right)}{3\sqrt{100-x^2}} \quad \text{.. as required}$$

iii. Find x when $\dfrac{dV}{dx} = 0$

♦♦ Mean mark 23%.

$$200 - 3x^2 = 0$$

$$x = \sqrt{\frac{200}{3}}$$

When $x < \sqrt{\dfrac{200}{3}}$, $\left(200-3x^2\right) > 0$

$$\Rightarrow \frac{dV}{dx} > 0$$

When $x > \sqrt{\dfrac{200}{3}}$, $\left(200-3x^2\right) < 0$

$$\Rightarrow \frac{dV}{dx} < 0$$

$\therefore$ MAX when $x = \sqrt{\dfrac{200}{3}}$

Equating the arc length of the section to the circumference of the cone:

$$2\pi r \cdot \frac{\theta}{2\pi} = 2\pi \cdot x$$

$$10\theta = 2\pi\sqrt{\frac{200}{3}}$$

$$\therefore \theta = \frac{2\pi \cdot 10\sqrt{2}}{10 \cdot \sqrt{3}}$$
$$= \frac{2\sqrt{2}\pi}{\sqrt{3}}$$

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8. Calculus, 2ADV C4 SM-Bank 12

i. $\text{Area} = \frac{1}{2} \times b \times h$

$$= \frac{1}{2}x\left(2e^{-\frac{x}{5}}\right)$$

$$\therefore A = xe^{-\frac{x}{5}}$$

ii. Stationary point when $\dfrac{dA}{dx} = 0,$

♦ *Mean mark (Vic) 35%.*

$$x\left(-\frac{1}{5}e^{-\frac{x}{5}}\right) + e^{-\frac{x}{5}} = 0$$

$$e^{-\frac{x}{5}}\left(1 - \frac{x}{5}\right) = 0$$

$$\therefore x = 5 \quad \left(e^{-\frac{x}{5}} > 0, \quad \text{for all } x\right)$$

When $x = 5, \quad A = xe^{-\frac{x}{5}} = 5e^{-1}$

$$\therefore A_{\text{max}} = \frac{5}{e} \text{ u}^2, \quad \text{when} \quad x = 5$$

iii. Find $S\colon F(0) = 2$

$$\Rightarrow S(0, 2)$$

♦♦ *Mean mark (Vic) 32%.*

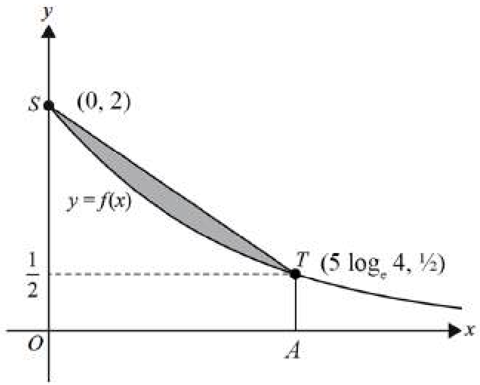
Find $T\colon \quad 2e^{-\frac{x}{5}} = \frac{1}{2}$

$$e^{-\frac{x}{5}} = \frac{1}{4}$$

$$-\frac{x}{5} = \log_e\left(\frac{1}{4}\right)$$

$$x = 5\log_e(4)$$

$$\Rightarrow T\left(5\log_e(4), \frac{1}{2}\right)$$



$$\therefore \text{ Area } = \text{Area } SOAT - \int_0^{5\log_e(4)} \left(2e^{-\frac{x}{5}}\right) dx$$

$$= \frac{1}{2}h(a+b) + \ 10\left[e^{-\frac{x}{5}}\right]_0^{5\log_e(4)}$$

$$= \frac{5}{2}\log_e(4)\left(2 + \frac{1}{2}\right) \ + 10\left[e^{-\log_e(4)} - e^0\right]$$

$$= \frac{25}{4}\log_e(4) + 10\left(\frac{1}{4} - 1\right)$$

$$= \frac{25}{4}\log_e(4) - \frac{15}{2} \ \text{u}^2$$

9. Trigonometry, 2ADV T3 SM-Bank 16

i. $h_{\min} = 65 - 55 \qquad h_{\max} = 65 + 55$

$\qquad = 10 \text{ m} \qquad\qquad = 120 \text{ m}$

ii. $\text{Period} = \dfrac{2\pi}{\frac{\pi}{15}} = 30 \text{ min}$

iii. $h\prime(t) = 65 - 55\cos\left(\dfrac{\pi t}{15}\right)$

$$h\prime(t) = \frac{\pi}{15} \times 55\sin\left(\frac{\pi}{15}t\right)$$

$$= \frac{11\pi}{3} \sin\left(\frac{\pi}{15}t\right)$$

Since $\sin\left(\dfrac{\pi}{15}t\right)_{\max} = \sin\left(\dfrac{\pi}{2}\right)$,

$\therefore h\prime(t)_{\max}$ occurs when

$$\frac{\pi t}{15} = \frac{\pi}{2}$$

$$\therefore t = \frac{\pi}{2} \times \frac{15}{\pi}$$

$$= \frac{15}{2} \text{ minutes} \quad (0 \le t \le 30)$$

10. Calculus, 2ADV C3 2019 HSC 15c

i. $\angle BRP = \angle PQA = 90^\circ \quad \text{(given)}$

$\angle BPR = \angle PAQ \quad \text{(corresponding)}$

$\Rightarrow \Delta\ BRP\ |||\ \Delta PQA \quad \text{(equiangular)}$

$$\frac{BR}{RP} = \frac{PQ}{QA} \quad \text{(corresponding sides of similar triangles)}$$

$$\frac{BR}{8} = \frac{1}{x}$$

$$BR = \frac{8}{x}$$

Using Pythagoras,

$$D^2 = (QA + RP)^2 + (BR + PQ)^2$$

$$= (x+8)^2 + \left(\frac{8}{x} + 1\right)^2$$

♦ *Mean mark part (i) 41%.*

ii. $(D^2)\prime = 2 \cdot (x+8) + 2(-1) \cdot \dfrac{8}{x^2}\left(\dfrac{8}{x} + 1\right)$

$$= 2x + 16 - \frac{16}{x^2}\left(\frac{8}{x} + 1\right)$$

$$= 2x + 16 - \frac{128}{x^3} - \frac{16}{x^2}$$

$$(D^2)\prime\prime = 2 - (-3) \cdot \frac{128}{x^4} - (-2) \cdot \frac{16}{x^3}$$

$$= 2 + \frac{384}{x^4} + \frac{32}{x^3}$$

When $x = 2$,

$$(D^2)\prime = 2 \times 2 + 16 - \frac{128}{8} - \frac{16}{4} = 0$$

$\Rightarrow$ S.P. at $x = 2$

$$(D^2)\prime\prime = 2 + \frac{384}{16} + \frac{32}{8} > 0$$

$\therefore$ MIN value of D^2 at $x = 2$.

♦♦ *Mean mark part (ii) 31%.*